

M.S. Student, Biomedical Engineering
Brain-Computer Interface Lab (Prof. Sung-Phil Kim)
Ulsan National Institute of Science and Technology (UNIST)

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RESEARCH INTERESTS

Brain-Computer / Brain-Machine Interfaces · Neural Encoding & Decoding of Vision and Motor Control · Cognitive Neuroscience of Sensory Perception · Brain Imaging

SKILLS

TOOLS MATLAB · Python · EEGLAB · SPM · FSL · MySQL

METHODS EEG · fMRI · Eye-tracking

EDUCATION

2025 – Current	M.S. in Biomedical Engineering Brain-Computer Interface Lab (Advisor: Prof. Sung-Phil Kim) · UNIST, Ulsan, South Korea
2019 – 2025	B.S. in Biomedical Engineering Minor in Electrical Engineering · UNIST, Ulsan, South Korea
2023 – 2024	Republic of Korea Air Force – Tactical Air Control Party Air Support Operations Center · Mandatory Military Service
2016 – 2018	Hansung Science High School Seoul, South Korea

RESEARCH EXPERIENCE

2020 – Current	Brain-Computer Interface Lab, UNIST Advisor: Prof. Sung-Phil Kim - Led human neurophysiology studies using EEG (and EEG + eye-tracking) with end-to-end responsibility from experimental design to analysis (MATLAB / EEGLAB; Python as needed). - Key contributions across projects: paradigm development and implementation, participant recruitment and session running, preprocessing / feature extraction, and interpretation / reporting of results.
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SELECTED PROJECTS

- EEG responses of emotion across OTT subscription fees** (*joint project with Korea University*) — Designed and implemented the experimental paradigm; conducted EEG experiments with 10+ participants; analyzed EEG data to characterize emotional differences across cost conditions.
- Neural basis of the choice-overload effect** — Developed and executed a complex EEG + eye-tracking paradigm; conducted experiments with 50+ participants; processed and analyzed large-scale datasets to probe neural processes under different preference conditions.
- fMRI study on tactile illusion** — Conducted 3T fMRI experiments with 30+ participants; analyzed BOLD signals using MATLAB / SPM / FSL; compared EEG vs. fMRI methodologies for sensory-mechanism studies.
- EEG signal analysis** — Signal processing and analysis using EEGLAB and MATLAB.

Spring 2022

Decision Neuroscience & Cognitive Engineering Lab, UNIST

Advisor: Prof. Dongil Chung

- **Resting-state EEG analysis of drug-induced parkinsonism and Parkinson's disease** (*joint project with Dongtan Sacred Heart Hospital*) — Analyzed resting-state EEG from PD and DIP patients; performed spectral analyses and EEG-clinical correlation analyses to identify differentiating neural signatures and support diagnostic insight.

Summer 2020

Bio and Neural Interfaces Lab, KAIST

Advisor: Prof. Seongjun Park

- **Mechanical interaction of thermally drawn fiber implants with brain tissue** — Investigated tissue-device interactions for fiber-based neural implants using in vivo surgery, histology, and computational modeling. Performed mouse-brain implant surgeries with fiber-based neural probes; verified immune responses via immunohistochemistry; conducted finite-element analysis to model probe-tissue mechanics.

PUBLICATIONS

- 2026 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Reducing cognitive demand fails to mitigate choice overload: multimodal neural and gaze evidence for effort conservation*. **iScience**.
UNDER 1ST REVISION
- 2026 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Effects of cognitive demand reduction on choice overload*. **bioRxiv**, 2026-02. PREPRINT
- 2024 **S. Seo**, S. Kim, S.-P. Kim, J.-H. Kim, S. Y. Kang & D. Chung. *Low-frequency EEG power and coherence differ between drug-induced parkinsonism and Parkinson's disease*. **Clinical Neurophysiology**, 168, 131–138.
- 2023 J. Park, S. Wang, S. Lee, **S. Seo**, N. Lee & S. Kim. *Viewer emotional response to webtoon-based drama: an EEG analysis*. **International Journal of Human-Computer Interaction**, 1–15.
- 2021 K. Kim, C. Sung, J. Lee, J. Won, W. Jeon, **S. Seo**, K. Yoon & S. Park. *Computational and histological analyses for investigating mechanical interaction of thermally drawn fiber implants with brain tissue*. **Micromachines**, 12(4), 394.
- 2021 S. Lee, **S. Seo**, N. Lee & S.-P. Kim. *The effect of OTT subscription fee on content engagement*. **The Journal of Korean Institute of Communications and Information Sciences**, 46(3), 572–582.

CONFERENCE PRESENTATIONS

- Nov 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *Closed-loop motor-imagery brain-computer interface training reshapes visual-motion perception*. **Society for Neuroscience (SfN) 2026**, Washington, DC.
- Jul 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric-psychometric dissociation in visual motion perception*. **Brain Engineering Society of Korea 2026**, Jeju, South Korea.
- May 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric function mirrors psychometric performance in visual motion perception*. **Vision Sciences Society (VSS) 2026**, St. Pete Beach, FL.
- Feb 2026 J. Kim, C.-E. Yoon, **S. Seo** & S.-P. Kim. *Real-time single-trial decoding in a calibration-free P300-based BCI using a compact eight-channel EEG configuration*. **14th International Conference on Brain-Computer Interface (BCI) 2026**, Jeongseon, South Korea.
- Sep 2024 **S. Seo**, S. Kim, S.-P. Kim, J. Kim, D. Chung & S. Y. Kang. *Quantitative EEG analysis for comparing drug-induced parkinsonism and Parkinson's disease*. **International Congress of Parkinson's Disease and Movement Disorders 2024**, Philadelphia, PA.

Nov 2022	J.-H. Kim, J. Kim, S. Seo & S.-P. Kim. <i>Somatosensory cortical representations of assimilation effects by vibrotactile stimulation: an fMRI study</i> . Society for Neuroscience (SfN) 2022 , San Diego, CA.
Sep 2022	S. Seo , S. Lee, N. Lee & S.-P. Kim. <i>Preference modulates the effect of choice overload with less involvement of working memory</i> . International Congress of Physiological Anthropology 2022 , Eugene, OR.
May 2022	S. Lee*, S. Seo* , N. Lee & S.-P. Kim. <i>Attenuated working memory accounts for choice overload in highly preferred alternatives</i> . 25th Annual Meeting of the Korean Society for Brain and Neural Sciences , Incheon, South Korea.

* Equal contribution.

RESEARCH GRANTS

2026.03 – Current	Augmentation Technology for Overcoming the Limits and Disorders of Human Cognition and Intelligence Graduate Researcher Information Technology and Telecommunications Planning & Evaluation Institute (IITP) · 2024.07 – 2031.12
2025.09 – Current	Music BrainCoder: A Non-Invasive BCI for Reproducing Imagined Music and Enhancing Musical Performance Graduate Researcher National Research Foundation of Korea (NRF) · 2025.03 – 2030.02
2022.08 – 2023.02	Future Brain Convergence Technology Development Project Undergraduate Researcher <i>"Development of brain-inspired computational models of active and adaptive intelligence for AI applications."</i> Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12
2022.04 – 2023.02	Brain Science Proprietary Technology Development Project Undergraduate Researcher <i>"Development of analysis methods for ultra-high-resolution neuroimaging signals and information-processing models of neural networks."</i> Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12
2022.03 – 2022.07	Smart Healthcare Research Center Undergraduate Researcher
2020.07 – 2022.02	Smart Media Services Innovation Convergence Project Undergraduate Researcher <i>"Development of neuro-artificial-intelligence convergence technology for next-generation OTT service."</i> Ministry of Science and ICT (MSIT) · 2020.07 – 2027.12

HONORS & AWARDS

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| 2022 | Excellent Research Award, 2nd Place
Department of Biomedical Engineering Faculty, UNIST, South Korea |
| 2021 | Student Creative Research Project, 1st Place
Smart Media Service Research Center, Korea University, South Korea |
| 2020 | Outstanding Grades for the Semester
Article 62 of the School Regulations · UNIST, South Korea |
| 2020 | Student Creative Research Project, 2nd Place
Smart Media Service Research Center, Korea University, South Korea |
| 2019 | Outstanding Grades for the Semester
Article 62 of the School Regulations · UNIST, South Korea |
| 2019 – Current | National Science & Engineering Undergraduate Scholarship
Korea Student Aid Foundation (Korean Government) · Full scholarship |

TEACHING EXPERIENCE

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| Spring 2026 | Machine Learning Methods for Neuroengineering
Teaching Assistant · UNIST |
| Fall 2025 | Brain & Cognitive Engineering
Teaching Assistant · UNIST |

SERVICES & OTHER PROFESSIONAL ACTIVITIES

PEER REVIEW

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| 2026 | Journal of Neural Engineering
Co-reviewer (with Prof. Sung-Phil Kim) · IOP Publishing |
| 2026 | Frontiers in Network Physiology
Reviewer (ad hoc) |

UNIVERSITY & DEPARTMENTAL SERVICE

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| 2022 | Undergraduate Student Council, UNIST
Election Campaign · Emergency Response Committee |
| 2021 | College of Information & Biotechnology Student Council, UNIST
Division of Academic Studies, School of Electrical Engineering |
| 2020 | School of Biomedical Engineering Student Council, UNIST
Preparation Task Force |
| 2020 | School of Electrical & Computer Science Engineering Student Council, UNIST
Emergency Response Committee · Planning Department |
| 2020 | Campus Assistant, Freshman Support Center, UNIST
Mentor for 18 first-year students (1 year) |
| 2020 | School of Human Factors Engineering Student Council, UNIST
Planning Department |